1 Directional Derivatives

We defined a multivariable function $f(x,y)$ to be differentiable at (a,b) if

$$\lim_{(x,y)\to(a,b)} \frac{f(x,y)-f(a,b)-f_x(a,b)(x-a)-f_y(a,b)(y-b)}{||(x,y)-(a,b)||}=0. \tag{1.1}$$

We saw last time that if f is differentiable at (a,b), then the partial derivatives $f_x(a,b)$ and $f_y(a,b)$ both exist at (a,b), which give the instantaneous rate of change of f at (a,b) in the $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$ directions, respectively. What if we want to know the instantaneous rate of change of f at (a,b) in some other direction?

Definition 1.2. Let $\hat{\mathbf{v}}=(v_1,v_2)$ be any unit vector. Define the *directional derivative* of f at (a,b) in the direction $\hat{\mathbf{v}}$, denoted by $D_{\hat{\mathbf{v}}}f(a,b)$, by

$$D_{\hat{\mathbf{v}}}f(a,b)=\lim_{t\to 0}\frac{f(a+tv_1,b+tv_2)-f(a,b)}{t},$$

provided this limit exists.

1. Write the parametric equations for the line parallel to the unit vector $\hat{\mathbf{v}}=(v_1,v_2)$ passing through the point (a,b).

2. Use your answer to the previous question to show that if f is differentiable at (a,b), then all of its directional derivatives exist.

3. In the previous question, you should have found that $D_{\hat{\mathbf{v}}}f(a,b)$ can be viewed as the dot product of a certain vector with $\hat{\mathbf{v}}$. Identify the components of this vector.

Definition 1.3 (Gradient vector)**.** Define the *gradient* vector of $f(x,y)$ at (a,b), denoted by $\nabla f(a,b)$, to be the vector of partial derivatives of f at (a,b):

$$\nabla f(a,b) = \begin{pmatrix} f_x(a,b) & f_y(a,b) \end{pmatrix}.$$

As you have just seen, the directional derivative $D_{\hat{\mathbf{v}}} f(a,b)$, if it exists, is found simply by taking the dot product of $\hat{\mathbf{v}}$ with the gradient of f at (a,b):

$$D_{\hat{\mathbf{v}}} f(a,b) = \nabla f(a,b) \cdot \hat{\mathbf{v}} = f_x(a,b)v_1 + f_y(a,b)v_2. \tag{1.4}$$

4. Calculate $D_{\hat{\mathbf{i}}} f(a,b)$ and $D_{\hat{\mathbf{j}}} f(a,b)$.

5. Let $f(x,y) = x^3 - 3xy + 4y^2$.

   (a) Calculate $\nabla f(x,y)$; that is, the gradient vector at an arbitrary point (x,y).

   (b) If $\hat{\mathbf{v}} = (\frac{\sqrt{3}}{2}, \frac{1}{2})$, find $D_{\hat{\mathbf{v}}} f(1,2)$. (To make your calculation simpler, evaluate ∇f at $(1,2)$ *before* taking the dot product.)

6. Let $f(x,y) = \sin x + e^{xy}$.

   (a) Calculate $\nabla f(x,y)$.

   (b) Calculate the directional derivative at the point $(2\pi, 1)$ in the direction of the vector $-4\hat{\mathbf{i}} + 8\hat{\mathbf{j}}$. (Careful! This is not a unit vector.)

The gradient and directional derivative is defined in the obvious way for a function of 3 or more variables: if $\hat{\mathbf{v}} = (v_1, v_2, v_3)$

$$D_{\hat{\mathbf{v}}} f(a,b,c) = \nabla f(a,b,c) \cdot \hat{\mathbf{v}} = f_x(a,b,c)v_1 + f_y(a,b,c)v_2 + f_z(a,b,c)v_3.$$

6. If $f(x,y,z) = x\sin(yz)$,

   (a) Calculate $\nabla f(x,y,z)$.

   (b) Calculate the directional derivative of f at $(1,3,0)$ in the direction of the vector $\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$.

2 Maximizing the Directional Derivative

7. Suppose f is a function of two or three variables and we consider all possible directional derivatives of f at a given point. These give the instantaneous rates of change of f in all possible directions.

   (a) In which direction does f change the fastest?

   (b) What is the maximum rate of change?

   Hint: Recall that one can write the dot product of two vectors $\mathbf{v}$ *and* $\mathbf{w}$ *as* $\mathbf{v}\cdot\mathbf{w} = ||\mathbf{v}||\,||\mathbf{w}||\cos\theta$, *where* $\theta \in [0,\pi]$ *is the angle between* $\mathbf{v}$ *and* $\mathbf{w}$.

8. Let $f(x,y) = xe^y$.

   (a) At the point $(2,0)$, in which direction does have have the maximum rate of change?

   (b) What is the maximum rate of change at $(2,0)$?

9. Suppose that the temperature at a point (x,y,z) in space is given by $T(x,y,z) = \frac{80}{1+x^2+2y^2+3z^2}$, where T is measured in degrees Celsius and x,y,z in meters. In what direction does the temperature increase fastest at the point $(1,1,-2)$? What is the maximum rate of increase?

10. If f is differentiable at (a,b), what is the direction of maximum rate of *decrease* at (a,b)?

11. Find the direction in which $f(x,y) = x^4y - x^2y^3$ is decreasing fastest at $(2,-3)$. What is the maximum rate of decrease?

12. Find all points at which the direction of fasest change of the function $f(x,y) = x^2 + y^2 - 2x - 4y$ is $\hat{\mathbf{i}} + \hat{\mathbf{j}}$.

Optional

We have seen that if f is differentiable at (a,b), then all of its directional derivatives exist at (a,b). Is the converse true? We have seen previously that the existence of both partial derivatives at (a,b) is not enough to guarantee differentiability at (a,b). Maybe if we have *all* directional derivatives then f will be differentiable at (a,b)?

Define $f : \mathbb{R}^2 \to \mathbb{R}$ by

$$f(x,y) = \begin{cases} \frac{x^3}{x^2+y^2}, & \text{if } (x,y) \neq (0,0) \\ 0, & \text{if } (x,y) = (0,0). \end{cases}$$

(a) If $\hat{\mathbf{v}} = (v_1, v_2)$ is any unit vector, show that $D_{\hat{\mathbf{v}}} f(0,0)$ exists.

(b) Conclude from the form of $D_{\hat{\mathbf{v}}} f(0,0)$ that f is nevertheless not differentiable at the origin. This shows that existence of all directional derivatives at a point is still not sufficient to guarantee differentiability at that point (as we have seen, we need the partial derivatives to also be continuous on an open neighborhood of that point).